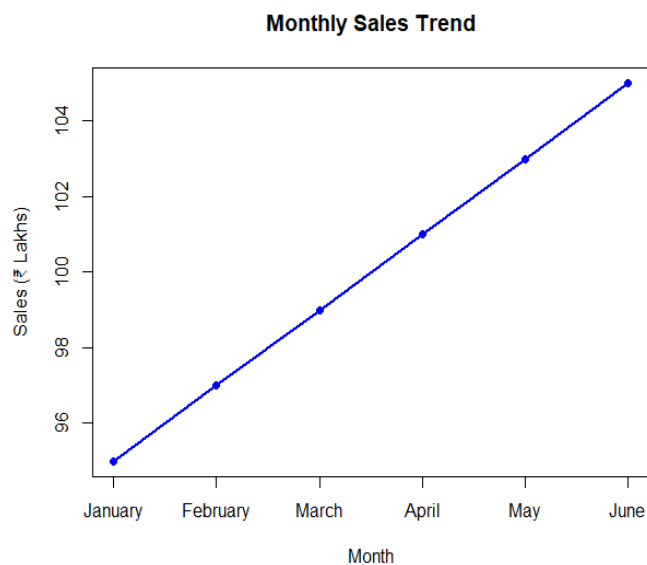


Question 1: Truncated Y-Axis in Sales Growth

```
# -----  
# Question 1: Truncated Y-Axis  
# -----  
# Dataset  
Month <- c("January", "February", "March", "April", "May", "June")  
Sales <- c(95, 97, 99, 101, 103, 105)  
# Create data frame  
sales_data <- data.frame(Month, Sales)  
  
# -----  
# Misleading Bar Chart  
# -----  
barplot(sales_data$Sales,  
        names.arg = sales_data$Month,  
        col = "tomato",  
        ylim = c(90, 110), # Truncated Y-axis  
        main = "Misleading Sales Growth (Y-axis starts at 90)",  
        xlab = "Month",  
        ylab = "Sales (₹ Lakhs)")  
# -----  
# Correct Bar Chart  
# -----  
barplot(sales_data$Sales,  
        names.arg = sales_data$Month,  
        col = "skyblue",  
        ylim = c(0, 110), # Correct Y-axis  
        main = "Correct Sales Growth (Y-axis starts at 0)",  
        xlab = "Month",  
        ylab = "Sales (₹ Lakhs)")
```

```
# -----
# Better Visualization (Line Chart)
# -----

plot(sales_data$Sales,
     type = "o",
     xaxt = "n",
     col = "blue",
     pch = 16,
     lwd = 2,
     xlab = "Month",
     ylab = "Sales (₹ Lakhs)",
     main = "Monthly Sales Trend")
axis(1, at = 1:6, labels = sales_data$Month)
```



1. Why is the chart misleading?

- The Y-axis starts at **90 instead of 0**.
- This exaggerates the difference between January (95) and June (105).
- The actual increase is only **10 lakhs (about 10.5%)**, but the chart makes it appear much larger.

2. Visualization principle violated

- **Honest representation of data (Integrity of visual encoding).**
- A bar chart should normally start the Y-axis at **0**, because bar length represents the value.

3. Recommended improved visualization

- Use a **bar chart with the Y-axis starting at 0**.
- Alternatively, use a **line chart**, which is more appropriate for showing changes over time.

4. How could this influence business decisions?

- Managers may believe sales have grown dramatically.
- They may increase production unnecessarily.
- They may allocate excessive marketing budgets.
- They may make incorrect inventory and investment decisions.
- This can lead to poor planning and financial losses.

Question 2: Pie Chart with Incorrect Proportions

```
# -----
```

```
# Question 2: Pie Chart
```

```
# -----
```

```
# Dataset
```

```
Department <- c("Computer Science", "Electronics", "Mechanical", "Civil")
```

```
Students <- c(420, 280, 200, 100)
```

```
# Calculate percentages
```

```
Percentage <- Students / sum(Students) * 100
```

```
# Display percentages
```

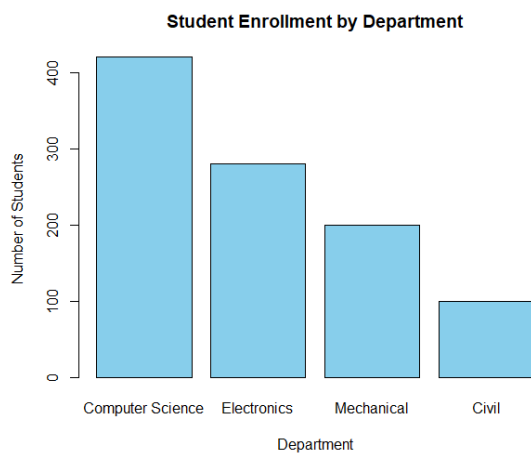
```
result <- data.frame(Department, Students, Percentage)
```

```
print(result)
```

```
# -----
```

```
# Correct Pie Chart
```

```
# -----
pie(Students,
    labels = paste(Department, round(Percentage,1), "%"),
    col = rainbow(4),
    main = "Correct Student Enrollment Distribution")
# -----
# Better Alternative: Bar Chart
# -----
barplot(Students,
    names.arg = Department,
    col = "skyblue",
    main = "Student Enrollment by Department",
    xlab = "Department",
    ylab = "Number of Students")
```



1. Calculate the correct percentage for each department

Total Students = 420 + 280 + 200 + 100 = 1000

Department	Students	Correct Percentage
Computer Science	420	42%

Department	Students	Correct Percentage
Electronics	280	28%
Mechanical	200	20%
Civil	100	10%

The correct pie chart should represent:

- **Computer Science – 42%**
- **Electronics – 28%**
- **Mechanical – 20%**
- **Civil – 10%**

2. Explain why the displayed chart is misleading

The displayed pie chart shows **50%, 30%, 15%, and 15%**, which do not match the actual student data.

This is misleading because:

- It **overestimates** Computer Science (50% instead of 42%).
- It **underestimates** Mechanical (15% instead of 20%).
- It **overestimates** Civil (15% instead of 10%).
- The chart gives viewers an incorrect understanding of the distribution of students across departments.
-

3. Suggest an alternative chart that would improve comparison accuracy

A **bar chart** is a better alternative because:

- It allows easy comparison of student counts.
- Differences between departments are clearer.
- It is easier to identify the largest and smallest departments accurately.
-

4. Discuss how such errors could affect institutional planning

An incorrect visualization can lead to poor decisions, such as:

- Allocating too many faculty members to the wrong department.
- Incorrect classroom and laboratory planning.
- Misallocation of budgets and resources.
- Wrong admission planning for future academic years.

- Poor strategic decisions based on inaccurate enrollment data.

Question 3: Unequal Bar Widths in Revenue Comparison

```
# -----
```

```
# Question 3: Revenue Comparison
```

```
# -----
```

```
# Dataset
```

```
Product <- c("A", "B", "C", "D")
```

```
Revenue <- c(60, 75, 90, 105)
```

```
# Create data frame
```

```
revenue_data <- data.frame(Product, Revenue)
```

```
# Display data
```

```
print(revenue_data)
```

```
# -----
```

```
# Correct Bar Chart
```

```
# -----
```

```
barplot(revenue_data$Revenue,
```

```
        names.arg = revenue_data$Product,
```

```
        col = "lightgreen",
```

```
        ylim = c(0, 120),
```

```
        main = "Annual Revenue Comparison",
```

```
        xlab = "Product",
```

```
        ylab = "Revenue (₹ Crores)")
```

```
# Add values on top of bars
```

```
text(x = c(0.7, 1.9, 3.1, 4.3),
```

```
     y = Revenue + 3,
```

```
     labels = Revenue)
```

```
# -----
```

```
# Better Alternative: Dot Plot
```

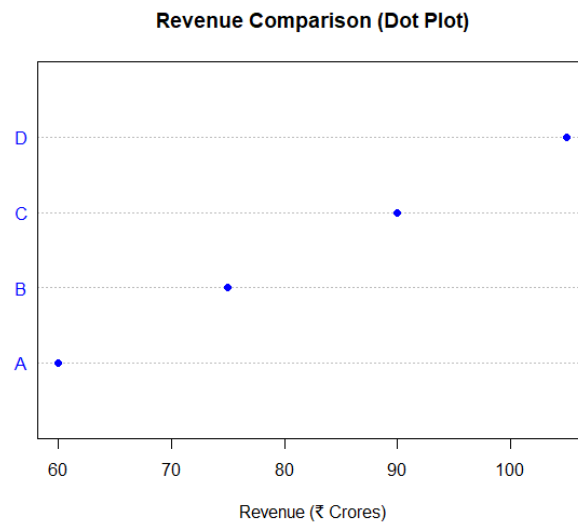
```
# -----
```

```
dotchart(revenue_data$Revenue,
```

```

labels = revenue_data$Product,
main = "Revenue Comparison (Dot Plot)",
xlab = "Revenue (₹ Crores)",
pch = 19,
col = "blue")

```



1. Identify the graphical error

The graphical error is the use of **unequal bar widths** in the bar chart.

In a bar chart:

- All bars should have the **same width**.
- Only the **height** of the bars should represent the data values.
-

2. Explain how unequal bar widths distort perception

When the widths of the bars are different:

- Larger-width bars appear more important because they occupy more area.
- Viewers compare both **height and width**, making Product D look much larger than it actually is.
- Although Product D's revenue is only **105 Crores** and Product A's is **60 Crores**, Product D may appear nearly **three times larger**, which is incorrect.
-

3. Redesign the visualization by specifying the correct design principles

A correct bar chart should follow these principles:

- Use **equal-width bars** for all products.

- Keep **equal spacing** between bars.
- Start the **Y-axis at 0**.
- Represent revenue **only by bar height**.
- Include clear axis labels, title, and data values if needed.

4. Discuss the possible consequences if investors rely on this chart

If investors believe the misleading chart:

- They may overestimate Product D's performance.
- They may invest more money based on incorrect information.
- They may ignore Products A, B, or C even if they perform well.
- Incorrect financial decisions could reduce investor confidence and damage the company's credibility.

Question 4: 3D Column Chart Distortion

```
# -----
```

```
# Question 4: Renewable Energy
```

```
# -----
```

```
# Dataset
```

```
Source <- c("Solar", "Wind", "Hydro", "Biomass")
```

```
Energy <- c(450, 500, 550, 520)
```

```
# Create data frame
```

```
energy_data <- data.frame(Source, Energy)
```

```
# Display dataset
```

```
print(energy_data)
```

```
# -----
```

```
# Recommended 2D Bar Chart
```

```
# -----
```

```
barplot(energy_data$Energy,
```

```
  names.arg = energy_data$Source,
```

```
  col = "lightblue",
```

```
  ylim = c(0, 600),
```

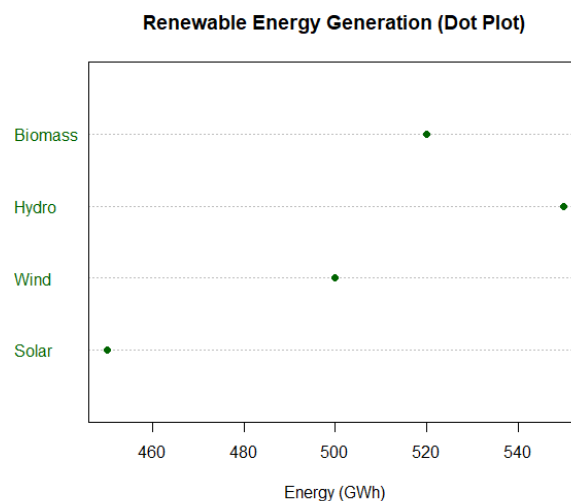
```
  main = "Renewable Energy Generation",
```



```

      xlab = "Energy Source",
      ylab = "Energy (GWh)")
# Add value labels
text(x = c(0.7, 1.9, 3.1, 4.3),
     y = Energy + 15,
     labels = Energy)
# -----
# Alternative: Dot Plot
# -----
dotchart(energy_data$Energy,
         labels = energy_data$Source,
         main = "Renewable Energy Generation (Dot Plot)",
         xlab = "Energy (GWh)",
         pch = 19,
         col = "darkgreen")

```



1. Explain why 3D charts can create misleading proportions

3D charts add perspective and depth, which can distort the size of bars.

This is misleading because:

- Bars in the front appear larger than those at the back.
- The viewer focuses on the **3D volume** instead of the actual height.
- It becomes difficult to compare the values accurately.

2. Compare the perceived difference with the actual difference

Source	Energy (GWh)
Solar	450
Wind	500
Hydro	550
Biomass	520

- **Actual difference (Hydro vs Wind) = $550 - 500 = 50$ GWh**
- **Percentage difference = $(50 / 500) \times 100 = 10\%$**

However, because of the 3D effect:

- Hydro appears about **40% larger** than Wind.
- This exaggerates the actual difference by making Hydro look much more productive than it really is.

3. Recommend a better visualization

A **2D bar chart** is the best choice because:

- All bars have the same width.
- Only the height represents the energy generated.
- It allows accurate comparison between energy sources.
- Avoid unnecessary 3D effects and perspective.

4. Discuss the importance of avoiding visual distortion in sustainability reporting

Accurate visualizations are important because:

- Governments and policymakers use these reports for planning.
- Investors rely on them when funding renewable energy projects.
- Misleading charts can result in incorrect resource allocation.
- Honest reporting improves transparency, credibility, and public trust.
- Correct visualizations support better environmental and business decisions.

Question 5: Bubble Chart with Incorrect Scaling

```
# -----
# Question 5: Disease Cases
# -----
# Dataset
District <- c("A", "B", "C", "D", "E")
Cases <- c(150, 250, 350, 450, 550)
# Create data frame
health_data <- data.frame(District, Cases)
# Display data
print(health_data)
```

```
# -----
# Incorrect Bubble Chart
# Radius directly proportional to cases
# -----
symbols(1:5, rep(1,5),
        circles = Cases/100,
        inches = 0.5,
        fg = "red",
        bg = "pink",
        xlab = "District",
        ylab = "",
        xaxt = "n",
        main = "Misleading Bubble Chart")
axis(1, at = 1:5, labels = District)
```

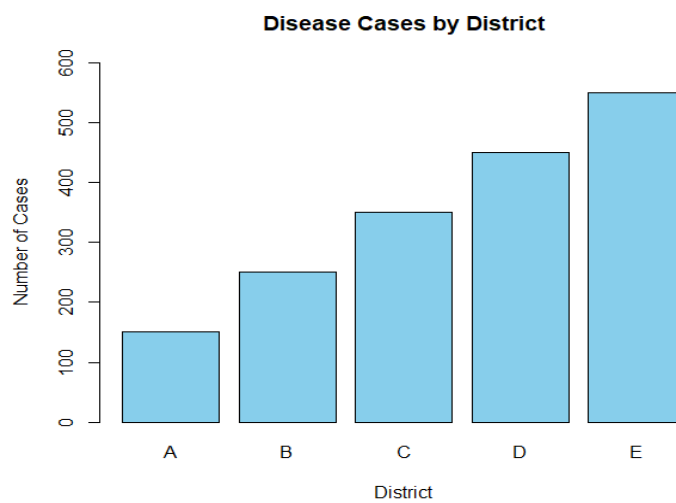
```
# -----
# Correct Bubble Chart
# Area proportional to cases
# -----
symbols(1:5, rep(1,5),
```

```

circles = sqrt(Cases),
inches = 0.1,
fg = "blue",
bg = "lightblue",
xlab = "District",
ylab = "",
xaxt = "n",
main = "Correct Bubble Chart")

axis(1, at = 1:5, labels = District)
# -----
# Alternative: Bar Chart
# -----
barplot(Cases,
        names.arg = District,
        col = "skyblue",
        ylim = c(0,600),
        main = "Disease Cases by District",
        xlab = "District",
        ylab = "Number of
Cases")

```



1. Explain why scaling the radius instead of the area is misleading

In a bubble chart, the **size of the bubble is perceived by its area**, not its radius.

If the radius is directly increased according to the number of cases:

- The bubble area increases by the square of the radius.
- Larger values become visually exaggerated.
- Districts with more cases appear much larger than their actual difference.

Example:

- District A = 150 cases
- District E = 550 cases

Actual ratio:

$$\frac{550}{150} = 3.67$$

So, District E has about **3.7 times more cases** than District A.

But when radius is scaled directly:

- The area becomes proportional to radius².
- District E may appear **more than 10 times larger**, creating a false impression.

2. Describe the correct method for proportional bubble sizing

The correct method is:

- Scale the **area of the bubble** according to the data value.
- Radius should be calculated using:

$$\text{Radius} = \sqrt{\text{Value}}$$

This ensures that:

$$\text{Bubble Area} \propto \text{Number of Cases}$$

Therefore, a district with twice the cases will have twice the bubble area, not twice the radius.

3. Suggest an alternative visualization suitable for this dataset

A **bar chart** is more suitable because:

- It provides accurate comparison of disease cases.
- Differences between districts are easier to understand.
- It avoids problems caused by incorrect bubble perception.

A sorted horizontal bar chart can clearly show which districts require more attention.

4. Discuss the potential impact on public health resource allocation

A misleading bubble chart can cause:

- Incorrect estimation of disease severity in districts.
- Over-allocation of medical resources to some areas.
- Neglect of districts that actually need attention.
- Wrong decisions regarding hospitals, healthcare workers, and medicine supply.
- Poor planning during disease control programs.

Accurate visualization helps authorities distribute resources fairly and effectively.